

Target: segment-level rates
for a digital/AI service item

No same-wave validation
possible: where a prior wave
fielded the item, use it as the
estimator (6.7 pp here, better
than any transferred correction);
elsewhere treat synthetic
estimates as hypotheses and do
not transfer a correction across
waves without re-benchmarking

no

Any *same-wave* real
probability-survey
data on the item?

yes

Segment entirely unob-
served *and* calibration
beats a real-only regres-
sion on held-out cells?

no

Real calibration sample large
(≥ 860 respondents for Gem-
ini, $\geq 1,730$ for EXAONE)?

yes

Estimate from the real data
alone, direct or partially pooled
(this study: 3.6 pp direct and
3.5 pp pooled at $n=2,598$;
the pooled version was
better at every tested size)

no

Real calibration sample
small (≤ 170 respondents for
Gemini, ≤ 430 for EXAONE)?

no

Undetermined zone (between
the two thresholds above; not
resolved by the tested sizes):
benchmark the calibrated panel
against direct estimation *and*
a partially pooled real-only
estimator on held-out splits,
and keep whichever wins

yes

yes

Model's bias signature
known and correctable
(e.g., age-structured)?

no / unknown

Do not use the synthetic
panel for estimation;
diagnostic use only (fram-
ing, consistency checks)

yes

Calibrated synthetic panel,
correction form selected
inside the calibration set
(6–10 pp at $n \approx 80$ –430; mar-
gin over a real-only shrink-
age estimator ≤ 2.2 pp,
and < 1 pp from $n \approx 170$);
always report the
real-only comparison